Agent Control Plane Reference Architecture (ACP-RA)

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Executive Summary

Agentic autonomy is shifting the unit of output from *human-hours* to *agent-hours*. That shift can increase decision tempo and execution density beyond human relevance windows, especially in contested environments where connectivity is intermittent and deception is routine.

The **Agent Control Plane (ACP)** makes this shift **fieldable**. It standardizes agent identity, delegated authority, long-running execution, and mediated action through gateways. It also produces evidence and replay artifacts for continuous authorization and after-action reconstruction.

This paper presents ACP-RA as a DoD CIO style reference architecture: purpose, principles, technical positions, patterns, and vocabulary. It guides and constrains downstream solution architectures rather than prescribing a single implementation.

1 Scope and Non-Goals

In scope

- Enterprise, intelligence, and operational-support agents that plan, coordinate, and invoke tools/actions within bounded authority.
- Multi-agent ensembles (swarms) and their coordination, messaging, shared state, budgets, observability, and containment.
- Work-unit management for long-running and parallel agent execution (checkpointing, dependencies, cancellation, supervision).
- Policy-as-code, evaluation-as-gate, and evidence generation for continuous authorization.
- Cross-enclave federation and cross-domain handoffs (identity, context, and evidence).

Out of scope

- Tactical employment guidance for autonomous weapons.
- Authorization of use-of-force decisions.
- Any design that bypasses applicable weapon system autonomy policy (see DoDD 3000.09).

2 Problem Statement

The Department is moving toward intent-driven systems that plan and act through tools. These systems can run many tasks in parallel and for long periods. Without a control plane, speed becomes unmanaged risk: unclear authority, uncontrolled data movement, fragile tool integration, weak evidence, and poor containment.

ACP-RA defines the required control surfaces to make autonomy governable at scale.

3 Strategic Drivers

- **A posture of acceleration:** platforms must support rapid onboarding and change while staying governable and reversible.
- **Supervision at scale:** the core problem shifts from whether agents can do tasks to whether humans can supervise many long-running parallel tasks.
- **Contested and degraded environments:** deception, poisoning, cascade failures, and intermittent connectivity are baseline constraints.
- **Federation:** cross-Service, cross-agency, and cross-enclave interoperability is unavoidable; the ACP must be protocol-neutral.

4 Architectural Principles

1. Agents are NPEs, not apps.
2. Centralize policy decisions; distribute enforcement.
3. Default deny; allow by explicit trust scope.
4. The doing boundary is explicit and mediated.
5. Evidence is first-class.
6. Rollback and containment are capabilities.
7. Context engineering is governed data movement.
8. Evals and monitoring are gates.
9. Budgets are policy.
10. Tools and skills are a supply chain.

5 Conceptual View (OV-1)

- Runtime plane: agent runtimes executing plans and invoking tools.
- Control plane: identity, trust scopes, work units, policy, gateways, evaluation, evidence, observability, containment.

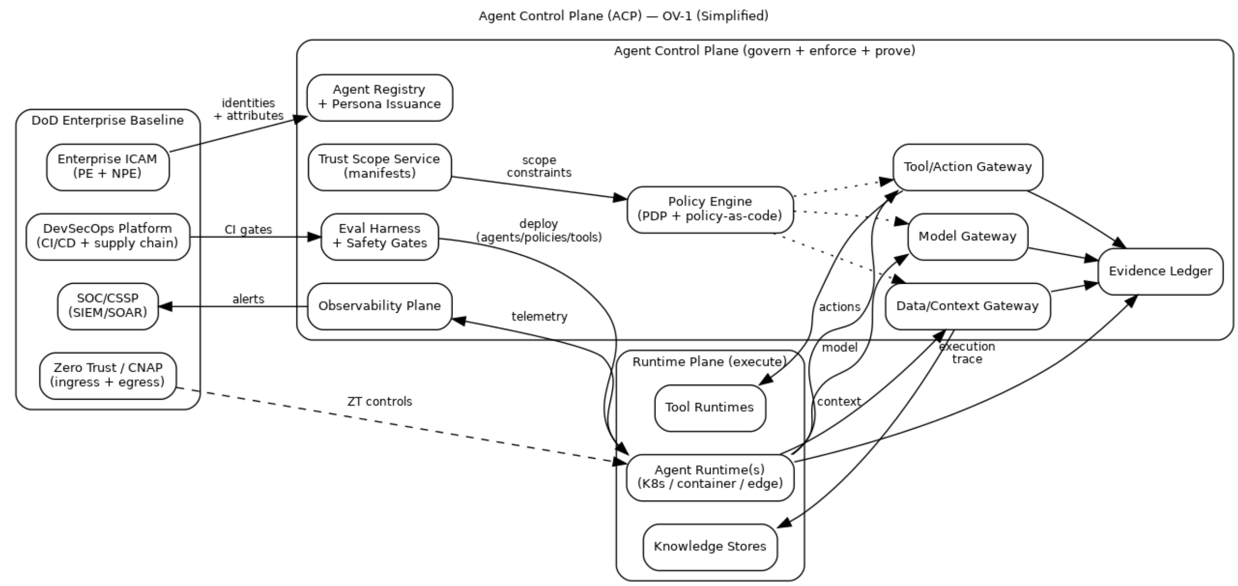

Figure 1: Agent Control Plane OV-1 (Simplified). The control plane governs identity, trust scopes, policy, gateways, evaluation, and evidence across the runtime plane. It aligns to enterprise identity and access management, DevSecOps, Zero Trust/CNAP, and security operations patterns.

6 Core Vocabulary

Agent: software actor that can plan, retrieve context, communicate, invoke tools, and execute actions within bounded authority.

Persona: controlled mission role for an agent; used by policy.

Trust scope manifest: signed, versioned contract defining delegated authority and constraints (tools/actions, consequence tier, environment, thresholds, budgets, evidence requirements, degraded modes).

Work unit: durable, supervised task thread for long-running and parallel execution that binds scope context, budgets, dependencies, checkpoint policy, sandbox provenance, and an evidence root.

Policy bundle: signed, versioned policy rules consumed by enforcement points.

Context bundle: replayable artifact representing retrieved context with provenance and integrity metadata and stable hash identifiers.

Action envelope: signed record of an attempted action/tool call including policy decision and evidence pointers.

Inter-agent message envelope: signed record of agent-to-agent messaging including schema, TTL, rate class, and provenance.

Ensemble (swarm): first-class multi-agent object with explicit governance, budgets, and containment semantics.

7 Consequence Tiers and Required Controls

Tier	Representative activities	Minimum controls
T0	Drafting, summarization, read-only analysis.	Agent identity, data tag controls, basic evidence, deny-by-default tool use.
T1	PR creation, ticket automation, recommendations (no direct apply).	Tool mediation, budgets, baseline evaluation, drift monitoring.
T2	Production changes, account/permission changes, external comms, GUI ops.	Human-on-the-loop approvals, dual control, enhanced evidence, rollback rehearsal.
T3	Life/safety or mission-critical actions.	Strict deny-by-default, quorum approvals, constrained degraded modes, full-fidelity evidence triggers.

Composition rule: when scopes combine, authority composes by intersection, never union.

8 Technical Positions (Required Control Surfaces)

- TP1: Agent identity as NPE (ICAM-aligned).
- TP2: Trust scopes are signed, versioned, and enforceable.
- TP3: Work units are first-class governance objects.
- TP4: Tool/action gateway mediates all doing.
- TP5: Tools and skills onboarded through supply-chain controls.
- TP6: Inter-agent gateway governs messaging and prevents cascades.
- TP7: Context/data gateway governs retrieval and context movement.
- TP8: Model gateway governs routing and upgrades.
- TP9: Evaluation harness gates promotion; monitoring gates runtime.
- TP10: Evidence ledger is tamper-evident and queryable.
- TP11: Degraded-mode behaviors are declared and enforced.

9 Architecture Components

This section defines bulkheads and gates:

- Agent registry and persona issuance.
- Trust scope service.
- Work unit service.

- Supervision console.
- Policy engine (policy decision point) with distributed enforcement points.
- Model gateway.
- Context and data gateway.
- Tool and action gateway.
- Inter-agent gateway.
- Evaluation harness.
- Evidence ledger and replay service.
- Observability.
- Containment and revocation.
- Budgets and resource governance.
- Federation services.

10 Control Loops

10.1 Action loop: intent → policy → mediated action → evidence

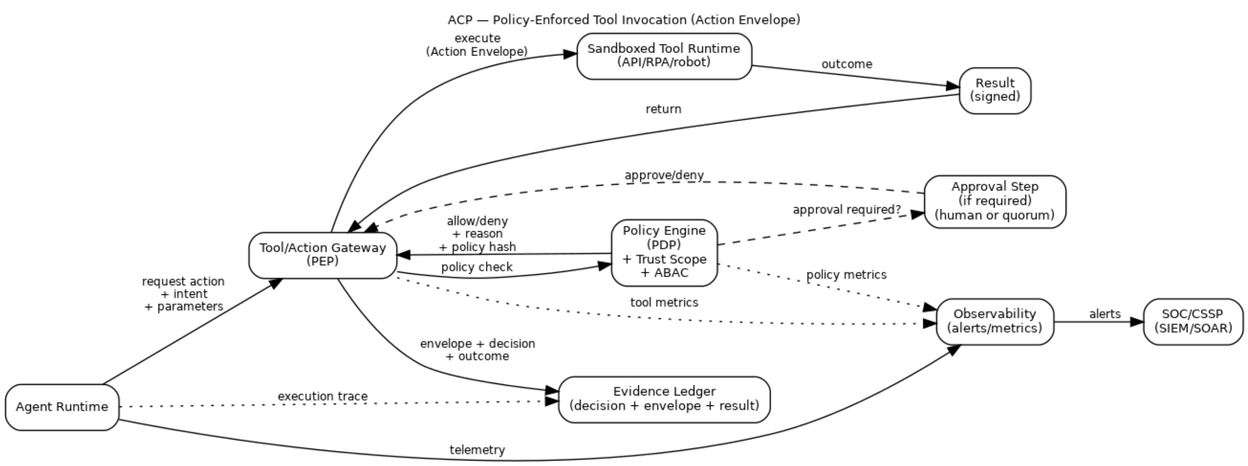

Figure 2: Policy-Enforced Tool Invocation (Action Envelope). The tool and action gateway acts as an enforcement point. It consults a policy decision point with the trust scope and attribute-based access control, optionally invokes approvals, emits tamper-evident evidence, and integrates with observability and security operations workflows.

10.2 Governance loop: artifacts → tests/evals → promotion → enforcement

- Policy, scope, tool catalog, and routing artifacts are versioned and signed.
- Evals and adversarial tests gate promotion.
- Promotion updates distributed enforcement points with rollback capability.

11 Multi-Agent Governance

Ensembles require explicit contracts: coordination pattern, arbitration rules, shared state governance, budgets and allocation, and containment without collapse.

12 Contested and Degraded Operation

ACP assumes poisoning, deception, and partial connectivity. Degraded modes are policy-driven safe behaviors (disconnected, intermittent, denied-model, human-handover).

13 Patterns

Work units as unit of supervision; policy-centered mesh; bulkhead gateways; budgeted autonomy; tool supply chain governance; evidence-first releases; ensemble contract.

14 Metrics and Roadmap

Define tempo, governance, swarm reliability, tool supply chain, adversarial robustness, and resource metrics. Maturity levels range from copilot sprawl to federated and contested operation.

15 Conclusion

ACP-RA treats autonomy as an engineered control system. It separates planning from doing, centralizes policy decisions while distributing enforcement, and makes evidence and containment first-class. This enables faster iteration without turning speed into unmanaged risk, including for multi-agent ensembles operating in contested and degraded environments.

16 References

References are maintained in the draft and will be normalized into a consistent bibliography pass.

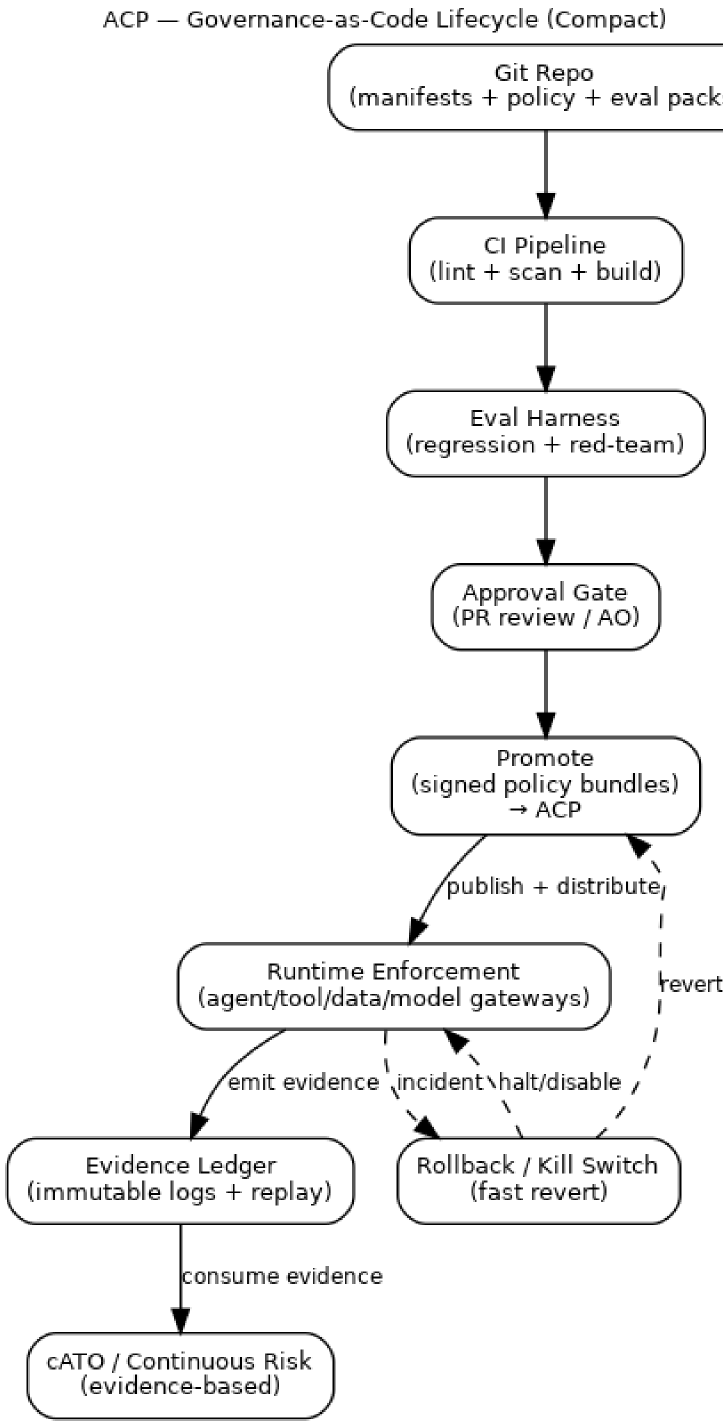

Figure 3: Governance-as-Code Lifecycle (Compact). Policy, manifests, and evaluation packs are versioned artifacts. The build pipeline and evaluation harnesses gate promotion of signed bundles to enforcement points, while evidence and cATO/continuous risk consume emitted telemetry and replayable logs.